
NASH BARGAINING PREFERENCE OPTIMIZATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Multi-objective alignment requires both a representation of preferences and a rule for selecting a compromise. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective through a regularized preference game and applies Nash bargaining to improvement over a common reference policy. A constrained KL-proximal formulation handles zero surplus at the reference; a prompt-separable finite-pool solver accounts for adaptive opponents; and pairwise log-ratio regression transfers its target to an LLM without evaluating a language-model normalizer. Exact population updates admit an $O(1/T)$ last-iterate guarantee. Controlled experiments support adaptive game values as preference nontransitivity increases. In an 8B model, short-schedule regression reduces held-out target error and finite-pool forward KL, with positive teacher-defined surpluses and near-base instruction-following and mathematics point estimates. Independent evaluation and repeated policy seeds remain necessary: the current natural panel does not establish an aggregation advantage or empirical Pareto optimality.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a standard approach for aligning large language models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard RLHF learns a scalar reward from pairwise comparisons. Direct Preference Optimization removes the separately trained reward model, but still assumes that comparisons are generated by an implicit scalar reward through a Bradley–Terry (BT) model (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches therefore require every response to admit one scalar score whose ordering explains all pairwise preferences.

This assumption can fail even within a single alignment objective. Human judgments can reflect several latent attributes whose relative importance changes with the prompt and the pair being compared (Tversky, 1969; Wang et al., 2024). As an illustrative example, consider three answers to a medical-advice question. Response A is concise and actionable but omits an important caveat; response B is comprehensive but verbose; and response C is less complete but includes the critical safety warning. Annotators may prefer A to B for relevance, B to C for completeness, and C to A for safety and reliability. These comparisons are cyclic: $A \succ B$, $B \succ C$, and $C \succ A$. Fitting the first two comparisons with a scalar reward requires $r(A) > r(B) > r(C)$, which contradicts the third. As a result, optimizing the fitted scalar reward can favor a response that appears best under the learned ranking but is still beaten by another response according to the original pairwise judgments (Munos et al., 2024).

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a second problem: after representing each objective without imposing transitivity, which compromise should one deployed policy realize? Scalar multi-objective methods expose this trade-off through weights, priorities, constraints, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). Recent game-theoretic methods preserve general pairwise feedback but combine objectives through Blackwell target sets or a worst-objective criterion (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026).

Classical Nash bargaining offers a priority-free way to select one compromise. It maximizes the product of each objective’s improvement over a common fallback (Nash, 1950). In alignment, the reference policy provides a natural fallback: an aligned policy should improve each objective relative to the model that would otherwise be deployed. The classical Nash solution is Pareto optimal and invariant to independent changes in the units used for different objectives. Prior alignment work has applied Nash welfare to learned scalar rewards, while game-based preference methods have focused mainly on minimax or Blackwell solutions (Zhong et al., 2024a; Navon et al., 2022; Zhang et al., 2026). What remains is a practical method for Nash bargaining when every objective value is itself defined by a policy-dependent preference game rather than by a fixed scalar reward.

We introduce Nash Bargaining Preference Optimization (NBPO) for this setting. Within each objective, NBPO evaluates the learner against an adaptive opponent that emphasizes responses exposing the learner’s weaknesses, while a KL penalty keeps the opponent near the reference policy. Across objectives, it maximizes the Nash product of the resulting improvements over that same reference. The classical bargaining properties matter here: at the population optimum no objective falls below the fallback, the outcome is Pareto efficient, and judge-specific units do not move the compromise.

The main technical challenge is optimization, not re-deriving the classical bargaining axioms. Three issues arise simultaneously: the reference policy has zero improvement and lies on the boundary of the logarithmic objective; each adaptive opponent depends on the policy being optimized; and an LLM cannot directly realize the normalized finite-action policy update. We address them with a constrained KL-proximal formulation, a low-dimensional dual whose weights are inversely proportional to achieved improvements, a prompt-separable direct solve of the finite-pool inner problem, and a partition-free pairwise log-ratio regression objective for projecting the finite-pool target into the language model.

Our contributions are:

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, separating within-objective preference representation from across-objective compromise selection.
- We derive a practical optimizer that handles the zero-surplus boundary and policy-dependent opponents. For each dual iterate, the sampled finite-pool problem separates into small strictly concave prompt-wise programs; their target policy is expressed for the LLM as a pairwise log-ratio regression objective, without fitting a scalar reward or evaluating a language-model partition function. We evaluate both held-out target fit and downstream generation.
- We prove $O(1/T)$ last-iterate convergence for exact population proximal updates. Feasibility-preserving controlled experiments isolate the benefit of adaptive game values under increasing nontransitivity, while a human-labeled SAFERLHF panel evaluates neural transfer, objective-wise outcomes, and capability retention. Released natural datasets are not treated as evidence of within-objective cycles unless their comparison graphs identify such cycles.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y | x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z | x) \in [0, 1]$ satisfying

$$P_k(y \succ z | x) + P_k(z \succ y | x) = 1, \quad P_k(y \succ y | x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z | x) = P_k(y \succ z | x) - \frac{1}{2}, \quad \Delta_k(y, z | x) = -\Delta_k(z, y | x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x), z \sim \nu(\cdot | x)} \Delta_k(y, z | x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex, and downward-closed feasible set with at least one strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G}: s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective's pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))\}. \quad (6)$$

The comparator minimizes the learner's pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z | x) := \mathbb{E}_{y \sim \pi(\cdot | x)} \Delta_k(y, z | x) \quad (7)$$

be the learner's expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k, \pi}^*(z | x) = \frac{\pi_{\text{ref}}(z | x) \exp\{-m_k^\pi(z | x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot | x)} \exp\{-m_k^\pi(z' | x)/\beta_k\}}, \quad (8)$$

$$V_{k, \beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot | x)} \exp\{-m_k^\pi(z | x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k, β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_{\pi} V_{k, \beta_k}(\pi)]_{x, y} = \mathcal{D}(x) Q_k^\pi(y | x), \quad Q_k^\pi(y | x) := \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot | x)} \Delta_k(y, z | x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^\pi(y | x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y | x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

The assumption states that at least one policy improves all objectives over the fallback. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_{\beta}(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_{\beta} := \{v \in \mathbb{R}^K : d \leq v \leq V_{\beta}(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_{\beta}$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_{\beta}, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. In LLM alignment, they mean that the population solution improves every objective over the reference, cannot be Pareto dominated by another feasible policy, and does not change merely because different judges report their values in different numerical units. Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective is undefined at initialization. Second, every game value contains an opponent that depends on the learner. Third, the population policy update must be realized by an autoregressive model. We handle these issues with a constrained proximal problem, a prompt-separable finite-pool inner solve, and neural log-ratio projection.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_{\pi} F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^{\pi}(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$.

The dual is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}, \quad (17)$$

where $\lambda = (\lambda_1, \dots, \lambda_K) \in \mathbb{R}_{++}^K$ is the vector of Lagrange multipliers. For fixed λ , let

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

The KL term makes the maximizing policy $\pi_t^*(\lambda)$ unique. Danskin’s theorem (Danskin, 1967) then allows us to differentiate the $\phi_t(\lambda)$ while holding this maximizing policy fixed, yielding

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the inverse-surplus weighting in (15) without selecting objective weights in advance.

5.2 DIRECT FINITE-POOL INNER SOLVE AND NEURAL PROJECTION

At one outer stage, sample learner candidates $\mathcal{Y}_x = \{y_{x,i}\}_{i=1}^{n_x}$ and comparator candidates $\mathcal{Z}_x = \{z_{x,j}\}_{j=1}^{m_x}$, and estimate the centered pairwise matrices $\hat{A}_{k,x}[i, j] = \hat{\Delta}_k(y_{x,i}, z_{x,j} | x)$. Let $p_{t,x} \in \Delta^{n_x}$ be the learner pool’s *occurrence measure* — the empirical importance weight each sampled candidate carries inside the finite program — and let μ_x be the corresponding occurrence measure on $\mathcal{Z}_x$. Both are counting measures over what was actually drawn, not the language model’s conditional law: with n_x i.i.d. draws from π_t and duplicates retained with multiplicity, $p_{t,x}(i) = 1/n_x$ regardless of the value of $\pi_t(y_{x,i} | x)$. The finite program below reweights that sampled support; the only bridge back to the model is the pool mapping $p_\theta(i) \propto p_{t,x}(i) \exp\{\log \pi_\theta(y_{x,i} | x) - \log \pi_t(y_{x,i} | x)\}$, which is what every finite-pool quantity we report is computed through. Reading $\text{softmax}_i \log \pi_\theta(y_{x,i} | x)$ as the pool distribution instead silently replaces the occurrence measure by a renormalized conditional and changes the sign of measured surpluses. For $p \in \Delta^{n_x}$, define

$$\hat{m}_{k,x}(p)_j := \sum_{i=1}^{n_x} p_i \hat{A}_{k,x}[i, j], \quad (21)$$

$$\hat{V}_{k,x}(p) := -\beta_k \log \sum_{j=1}^{m_x} \mu_{x,j} \exp\{-\hat{m}_{k,x}(p)_j / \beta_k\}. \quad (22)$$

The disagreement value $\hat{d}_k$ is measured from an independently constructed reference–reference tensor, not set to zero.

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , pairwise models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T .

```

1: Sample finite learner/comparator pools; construct pairwise and reference-reference tensors;
   estimate  $\hat{d}_{1:K}$ .
2: Set  $\pi_0 \leftarrow \pi_{\text{ref}}$ .
3: for  $t = 0, \dots, T - 1$  do
4:   Score  $\pi_t$  on the learner pool and initialize positive  $\lambda$ .
5:   repeat
6:     Solve (23) independently for every prompt.
7:     Aggregate  $\hat{s}_{1:K}$  and solve  $\hat{s}_k = 1/\lambda_k$  for  $\lambda$  over  $\Lambda$ .
8:   until the projected dual/KKT criteria are met
9:   Construct  $\hat{h}_t^*$  by (24) and verify (27).
10:  Fit one neural candidate  $\tilde{\pi}_{t+1}$  using (26).
11:  Set  $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$  if held-out empirical surpluses are positive; otherwise retain  $\pi_t$ .
12: end for
13: return  $\pi_T$ .
```

For fixed λ , the empirical counterpart of (18) separates across prompts:

$$p_{t,x}^*(\lambda) \in \arg \max_{p \in \Delta^{n_x}} \left\{ \sum_{k=1}^K \lambda_k \hat{V}_{k,x}(p) - \frac{1}{\eta_t} \text{KL}(p \| p_{t,x}) \right\}. \quad (23)$$

Each problem is strictly concave because of the KL term. We solve these small programs directly to a declared numerical tolerance and aggregate their values to obtain $\hat{s}_k(p_t^*(\lambda))$. Dual stationarity is the K -dimensional condition $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$, which we solve as a root problem in $\log \lambda$ over $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$; every reported run uses this solve and reports its projected KKT residual. A projected subgradient fallback on the same condition stalls three orders of magnitude short of that tolerance (Appendix A.5). In contrast, repeatedly alternating a closed-form opponent with the exponential policy map need not converge under strong nontransitivity; we retain that R -step map only as an ablation.

Once the dual converges, the finite-pool target defines the pairwise log-ratio change

$$\hat{h}_t^*(x, i, i') := \log \frac{p_{t,x}^*(i)}{p_{t,x}(i)} - \log \frac{p_{t,x}^*(i')}{p_{t,x}(i')}. \quad (24)$$

For a neural policy π_θ , let

$$h_t(\pi_\theta; x, y, y') := \log \frac{\pi_\theta(y \mid x)}{\pi_\theta(y' \mid x)} - \log \frac{\pi_\theta(y' \mid x)}{\pi_\theta(y \mid x)}. \quad (25)$$

We project the finite-pool target into the LLM with

$$\mathcal{L}_{\text{reg}}^{(t)}(\theta) = \mathbb{E}_{x, (i, i') \sim \xi_x} \left[h_t(\pi_\theta; x, y_{x,i}, y_{x,i'}) - \hat{h}_t^*(x, i, i') \right]^2. \quad (26)$$

No language-model partition function appears. At the inner optimum,

$$\hat{h}_t^*(x, i, i') = \eta_t \sum_{k=1}^K \lambda_k \left(\hat{Q}_{k,x}^*(i) - \hat{Q}_{k,x}^*(i') \right), \quad (27)$$

which provides an independent solver and artifact check. Pairwise probabilities may come from direct human comparisons or a general pairwise preference model; NBPO never requires them to decompose into scalar response rewards.

We compare (26) with weighted behavior cloning of the same finite-pool target. Appendix E defines this alternative realization and its matched pairwise implementation.

The finite-pool solve is numerical and the LLM projection remains approximate. The convergence result below concerns exact population solutions of (16).

Last-iterate convergence. Algorithm 1 returns the final outer policy rather than an average of past policies, so the relevant guarantee is on the last population iterate. Under Assumption 1, with $\pi_0 = \pi_{\text{ref}}$ and (16) solved exactly at every outer iteration with arbitrary $\eta_t > 0$, the Nash welfare is monotone, $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and $F(\pi^*) - F(\pi_T) \leq D(\pi^* \| \pi_{\text{ref}}) / \sum_{t < T} \eta_t$: a constant proximal step gives an $O(1/T)$ final-iterate gap. Theorem 1 and its proof are in Appendix A.6.

6 EXPERIMENTS

We evaluate preference representation, target realization, and generated-policy quality. Objective-wise outcomes and general capability are reported separately, following multi-objective alignment and NLHF (Agnihotri et al., 2026; Zhang et al., 2026; Munos et al., 2024).

6.1 CONTROLLED PREFERENCE NONTRANSITIVITY

Each instance has 40 prompts and four objectives. We add a skew-symmetric cyclic component to transitive payoffs while preserving positive joint improvement, with attainable worst surplus $\rho^* = \max_{\pi} \min_k s_k(\pi) \approx 0.03$. All methods are evaluated under the same adaptive-game values across five independent seeds. The matched controls vary preference representation or aggregation; Appendix B gives the construction and full results.

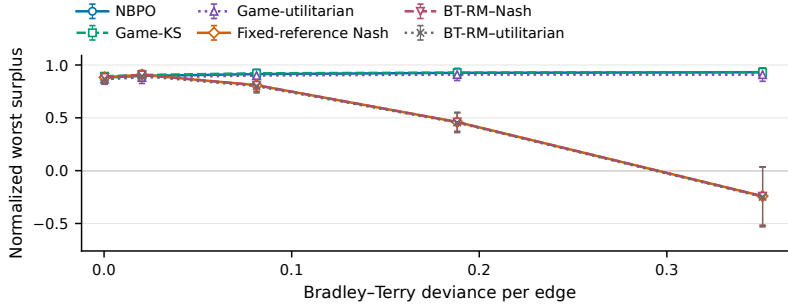


Figure 1: Controlled nontransitivity. Points show mean BT deviance and normalized worst surplus across five seeds; bars show one sample standard deviation. All six plotted methods converge. Full results and the legacy-map ablation are in Appendix B.

At the strongest perturbation, NBPO reaches 0.931 normalized worst surplus versus -0.241 for fixed-reference Nash and -0.242 for BT-RM-Nash. Game-KS and game-utlitarian also remain stable. This supports the representation claim; it does not establish Nash superiority over all aggregation rules. Independent solver residuals and robustness across constructions are reported in Appendix B.

6.2 OBJECTIVE-WISE ALIGNMENT AND GENERAL CAPABILITY

Setup. We fine-tune Llama-3.1-8B-Instruct on 2,000 SAFERLHF prompts using frozen, objective-specific preference-model ensembles. Development and held-out evaluation use 500 and 1,000 prompts. Both realizations share the solved target and response pools. The released human comparisons disagree across helpfulness and harmlessness on 24.33% of held-out pairs, but contain no fully observed triangles and cannot identify within-objective human cycles. Model-predicted disagreement on generated pools is a separate quantity (Appendix C).

Teacher-defined gains. MSE-250 attains worst-objective surplus 0.1186, versus 0.1337 for MSE-1750 and 0.0569 for WBC-250 (Table 1). These are evaluations by the training preference models, relative to a stochastic-reference disagreement point. Gains concentrate on prompts where the base gives a short refusal; joint improvement alone does not identify a Nash-specific effect. Teacher truncation and this post-hoc subgroup analysis are discussed in Appendix F.

Table 1: Teacher-defined objective evaluation on 1,000 held-out prompts. Surplus is $\hat{V}_k - \hat{d}_k$ using common frozen comparator pools. Greedy-decoder rows do not certify stochastic-policy acceptance. The finite-pool target is a reference, not an upper bound. MSE-250 objective entries are rounded to three decimals in the execution report; its minimum is reported separately. Dashes mark unmeasured comparisons.

Method	Updates	$s_{\text{help}} \uparrow$	$s_{\text{safe}} \uparrow$	$\min_k s_k \uparrow$
Finite-pool target p^*	—	+0.1653	+0.1662	+0.1653
Base (greedy)	—	−0.0405	−0.0320	−0.0405
NBPO-MSE (1750)	1750	+0.1339	+0.1337	+0.1337
NBPO-WBC (1750)	1750	+0.0738	+0.0808	+0.0738
NBPO-MSE (250) [†]	250	+0.119	+0.125	+0.1186
NBPO-WBC (250) [†]	250	—	—	+0.0569
Game-utilitarian	250	—	—	—
Fixed-reference Nash	250	—	—	—
BT-RM–Nash	250	—	—	—
PROSPER	—	—	—	—
MOPO	—	—	—	—
Game-maxmin	250	—	—	—
Scalarized DPO	—	—	—	—

Table 2: Capability and safety of the same checkpoints. Higher is better except harmful completions and benign refusals. Local reference-win proxies are not official AlpacaEval or Arena-Hard scores. Harm: HarmBench; refusal: XSTest safe prompts; tokens: Alpaca median. Measured rows currently use one seed; missing results are dashes.

Method	IFEval	GSM8K	Harm↓	Refusal↓	Alpaca	AH-hard	AH-creative	Tokens
Base (greedy)	.752	.867	.281	—	.500	—	—	—
NBPO-MSE (1750)	.758	.864	.284	—	.419	—	—	—
NBPO-WBC (1750)	.741	.852	.256	—	.578	—	—	—
NBPO-MSE (250) [†]	.760	.870	.269	—	.497	—	—	—
NBPO-WBC (250) [†]	.741	.867	.266	—	.486	—	—	—
Game-utilitarian	—	—	—	—	—	—	—	—
Fixed-reference Nash	—	—	—	—	—	—	—	—
BT-RM–Nash	—	—	—	—	—	—	—	—
PROSPER	—	—	—	—	—	—	—	—
MOPO	—	—	—	—	—	—	—	—
Game-maxmin	—	—	—	—	—	—	—	—
Scalarized DPO	—	—	—	—	—	—	—	—

Aggregation sensitivity. The Nash weights are nearly equal, $\lambda = (6.325, 6.186)$. Equal weights with the same ℓ_1 mass change the target by mean TV 0.0008 and pairwise-target RMS 0.0034 (Appendix E.1). This pool provides limited target-level separation between the two rules. Policy differences and training-seed variability have not been measured; a null policy comparison would not establish equivalence.

External evaluation. MSE-250 obtains IFEval 0.760, GSM8K 0.870, and harmful-completion rate 0.269, compared with 0.752, 0.867, and 0.281 for the base. Its Alpaca local win rate is 0.497; these point estimates do not establish equivalence or overall superiority. WBC-1750 has a higher local win rate (0.578) despite lower teacher-defined surplus. Independent objective evaluation, policy-seed repeats, and a scalarization sweep are required to assess empirical nondominance (Appendix F.1).

6.3 PROJECTION AND TRAINING SCHEDULE

We compare canonical regression with WBC under matched data and policy-token exposure. The original comparison uses 1750 updates. After inspecting its development trajectories, we declared separate 250-update follow-ups, each with its own cosine schedule. They are development-informed runs, not truncated long runs or a fully prospective 2×2 study.

Both short runs pass the existing target-fit thresholds: $\text{nMSE} < 0.90$, $\text{sign} > 0.65$, and positive Pearson and Spearman correlations. MSE-250 also reduces finite-pool KL from 0.7879 to 0.6204.

Table 3: Target realization on 500 development prompts, one seed. Each horizon has its own cosine schedule; 250-update runs are development-informed follow-ups. Pool KL is $D_{\text{KL}}(p^* || \hat{p}_\theta)$ on sampled support; the reference value is 0.7879. Full diagnostics are in Appendix E.

Projection	Horizon	nMSE	Sign	Pearson	Pool KL↓	GPU-h
MSE	1750	1.418	0.693	0.415	1.2131	7.48
WBC	1750	13.070	0.486	−0.152	6.0634	6.78
MSE†	250	0.690	0.702	0.588	0.6204	2.20
WBC†	250	0.780	0.687	0.492	0.6875	2.08

MSE-1750 has positive correlation but increases KL; it already fails the original nMSE threshold. Thus correlation alone is insufficient, while no gate false positive is demonstrated here. These are importance-reweighted pool distributions, not full-policy KLs. Update count, exposure, and learning-rate trajectory change together, so horizon is not isolated as the sole cause.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026). NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Reward-based multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize weakest-objective criteria, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These methods are important empirical baselines because they explicitly fit or use scalar objective rewards, whereas NBPO learns from general pairwise feedback without assuming that each objective is transitive.

Social choice and bargaining. The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2024a); Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

NBPO combines adaptive preference-game values with Nash bargaining over reference-relative improvements. A constrained proximal formulation yields a direct finite-pool solver and an $O(1/T)$ exact-population guarantee. Controlled experiments support adaptive game representations. Short-schedule 8B regression improves target fit and teacher-defined outcomes with near-base capability point estimates. An aggregation advantage and empirical Pareto optimality remain unestablished. Teacher truncation, refusal-concentrated gains, development-informed schedules, and a single policy seed limit the conclusions. Joint improvement need not be feasible, and population game values provide no per-prompt safety guarantee.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 distinguishes the population proximal problem, the numerically solved finite-pool inner problem, and its neural projection. Appendix A contains the dual and convergence proofs; Appendices B–F specify constructions, supervision, hyperparameters, diagnostics, and evaluation protocols. The submission package should include candidate pools, preference tensors, dual traces, solver residuals, per-seed outputs, and the code and configurations required to reproduce the reported results.

REFERENCES

- Akhil Agnihotri, Rahul Jain, Deepak Ramachandran, and Zheng Wen. Multi-objective preference optimization: Improving human alignment of generative models. In *International Conference on Machine Learning*, 2026.
- Kush Bhatia, Ashwin Pananjady, Peter L. Bartlett, Anca D. Dragan, and Martin J. Wainwright. Preference learning along multiple criteria: A game-theoretic perspective. In *Advances in Neural Information Processing Systems*, volume 33, pp. 7413–7424, 2020.
- David Blackwell. An analog of the minimax theorem for vector payoffs. *Pacific Journal of Mathematics*, 6(1):1–8, 1956.
- Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Daniele Calandriello, Daniel Guo, Rémi Munos, Mark Rowland, Yunhao Tang, Bernardo Avila Pires, Pierre Harvey Richemond, Charline Le Lan, Michal Valko, Tianqi Liu, et al. Human alignment of large language models through online preference optimisation. *arXiv preprint arXiv:2403.08635*, 2024.
- Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Vincent Conitzer, Rachel Freedman, Jobst Heitzig, Wesley H. Holliday, Bob M. Jacobs, Nathan Lambert, Milan Mossé, Eric Pacuit, Stuart Russell, Hailey Schoelkopf, Emanuel Tewelde, and William S. Zwicker. Position: Social choice should guide AI alignment in dealing with diverse human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- John M. Danskin. *The Theory of Max-Min and Its Application to Weapons Allocation Problems*. Springer-Verlag, 1967.

- Rémi Munos, Michal Valko, Daniele Calandriello, Mohammad Gheshlaghi Azar, Mark Rowland, Daniel Guo, Yunhao Tang, Matthieu Geist, Thomas Mesnard, Andrea Michi, et al. Nash learning from human feedback. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- John F Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.
- Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-task learning as a bargaining game. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2022.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pp. 27730–27744, 2022.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Alexandre Ramé, Guillaume Couairon, Corentin Dancette, Jean-Baptiste Gaya, Mustafa Shukor, Laure Soulier, and Matthieu Cord. Rewarded soups: Towards pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Corby Rosset, Ching-An Cheng, Arindam Mitra, Michael Santacrose, Ahmed Awadallah, and Tengyang Xie. Direct nash optimization: Teaching language models to self-improve with general preferences. *arXiv preprint arXiv:2404.03715*, 2024.
- Seongho Son, William Bankes, Sangwoong Yoon, Shyam Sundhar Ramesh, Xiaohang Tang, and Ilija Bogunovic. Robust multi-objective controlled decoding of large language models. In *International Conference on Learning Representations*, 2026.
- Amos Tversky. Intransitivity of preferences. *Psychological Review*, 76(1):31–48, 1969.
- Haoxiang Wang, Wei Xiong, Tengyang Xie, Han Zhao, and Tong Zhang. Interpretable preferences via multi-objective reward modeling and mixture-of-experts. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 10582–10592, 2024.
- Yue Wu, Zhiqing Sun, Huizhuo Yuan, Kaixuan Ji, Yiming Yang, and Quanquan Gu. Self-play preference optimization for language model alignment. In *International Conference on Learning Representations*, 2025.
- Jiahao Zhang, Lujing Zhang, Keltin Grimes, Zhuohao Yu, Gokul Swamy, and Zhiwei Steven Wu. Back to blackwell: Closing the loop on intransitivity in multi-objective preference fine-tuning. *arXiv preprint arXiv:2602.19041*, 2026.
- Yuheng Zhang, Dian Yu, Baolin Peng, Linfeng Song, Ye Tian, Mingyue Huo, Nan Jiang, Haitao Mi, and Dong Yu. Iterative nash policy optimization: Aligning LLMs with general preferences via no-regret learning. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=Pujt3ADZgI>.
- Huiying Zhong, Zhun Deng, Weijie J. Su, Zhiwei Steven Wu, and Linjun Zhang. Provable multi-party reinforcement learning with diverse human feedback. *arXiv preprint arXiv:2403.05006*, 2024a.
- Yifan Zhong, Chengdong Ma, Xiaoyuan Zhang, Ziran Yang, Haojun Chen, Qingfu Zhang, Siyuan Qi, and Yaodong Yang. Panacea: Pareto alignment via preference adaptation for LLMs. In *Advances in Neural Information Processing Systems*, 2024b.
- Runlong Zhou, Maryam Fazel, and Simon S. Du. Extragradient preference optimization (EGPO): Beyond last-iterate convergence for nash learning from human feedback. In *Conference on Language Modeling*, 2025.

Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin’s theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (28)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_{\nu} J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k, β_k} is continuous, $V_{\beta}(\pi)$ is bounded. The definition of $\mathcal{G}_{\beta}$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_{\beta}$, choosing policies π_n with $d \leq v_n \leq V_{\beta}(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_{\beta}$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_{\beta}(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_{\beta}(\pi_1) + (1 - \alpha)V_{\beta}(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_{\beta}$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_{\beta}$ with $v > d$, some policy π satisfies $v \leq V_{\beta}(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k, \beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_{\beta}(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_{\beta}, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatewise monotonicity implies Pareto optimality. Strict

concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

Table 4: Classical population guarantees of common aggregation rules under joint improvement. The Nash row follows from the standard bargaining solution; the other rows are supported by the finite counterexamples below.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

The positive guarantees in Table 4 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maxmin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maxmin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

For the uniform reference,

$$\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$$

and hence

$$d = (-1/3, -1/12).$$

First, joint improvement is feasible. The policy

$$\bar{\pi} = (1/4, 1/4, 1/2)$$

is only a feasibility witness; it gives

$$\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$$

Therefore

$$V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$$

In contrast, solving the equal-weight utilitarian problem $\max_{\pi} V_1(\pi) + V_2(\pi)$ gives

$$\pi_U = (0, 1/5, 4/5).$$

For this policy,

$$\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$$

so

$$V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$$

Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

Absolute-maxmin failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again,

$$d = (-1/3, -1/12).$$

Joint improvement is feasible, as witnessed by

$$\bar{\pi} = (1/3, 7/15, 1/5),$$

for which

$$V(\bar{\pi}) = (-4/15, -1/60)$$

and therefore

$$s(\bar{\pi}) = (1/15, 1/15) > 0.$$

Absolute maxmin instead solves

$$\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$$

whose linear-program solution is

$$\pi_{AM} = (0, 4/5, 1/5).$$

For this policy,

$$\pi_{AM}^\top \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{AM}^\top \Delta_2 = (-1/10, -1/10, 2/5),$$

so

$$V(\pi_{AM}) = (-1/10, -1/10)$$

and

$$s(\pi_{AM}) = (7/30, -1/60).$$

Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed $1/6$.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at $1/6$. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION AND REGRESSION FIXED POINT

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin's theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Dual step used as a fallback. Stationarity $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$ can also be approached by projected subgradient *descent* on the dual — the update below subtracts the dual gradient, and the dual of the concave inner problem is minimized over λ —

$$\lambda^{(m+1)} = \Pi_\Lambda \left[\lambda^{(m)} - \gamma_m \left(\hat{s} \left(p_t^*(\lambda^{(m)}) \right) - (\lambda^{(m)})^{-1} \right) \right], \quad (29)$$

where $\gamma_m > 0$ is the dual step size, Π_Λ is Euclidean projection onto $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ with $0 < \lambda_{\min} < \lambda_{\max}$, and the inverse is elementwise. On the pools reported here this iteration stalls at a projected KKT residual near 10^{-2} after 3000 steps, whereas the root solve reaches 10^{-9} in tens of inner solves; no reported result uses (29).

What the projected certificate certifies. Both the root solve and the fallback act on the box $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$, so the residual we report is the projected KKT residual of the *box-constrained* dual. That residual certifies the original problem only when the box is inactive at the returned λ : on an active bound the projection can annihilate a genuine stationarity violation, and a small number then certifies nothing. We therefore report bound activity alongside every residual and treat a run with an active bound as uncertified rather than converged. In the runs reported here the multipliers are interior by a wide margin — on the SAFERLHF pool, $\Lambda = [10^{-3}, 10^3]^2$ and the solved $\lambda = (6.33, 6.19)$ sit at least two decades from either bound in $\log_{10} \lambda$ — and the stationarity residual recomputed independently from the returned policy is 6.7×10^{-11} .

Finite-pool optimality and neural projection. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (30)$$

The direct finite-pool program (23) is the sampled analogue of this strictly concave problem. Its normalized optimizer determines all finite-pool log-probability differences, so (24) is identified without computing a language-model normalizer. Equation (27) is the corresponding finite-pool KKT identity and is checked numerically before an artifact is accepted. With a full-support pair proposal, zero regression error implies

$$\tilde{p}_{\theta, x}(i) = \frac{p_{t, x}(i) \pi_{\theta}(y_{x, i} | x) / \pi_t(y_{x, i} | x)}{\sum_j p_{t, x}(j) \pi_{\theta}(y_{x, j} | x) / \pi_t(y_{x, j} | x)} = p_{t, x}^*(i).$$

This is an importance-reweighted occurrence measure, not the raw LM probabilities normalized within the pool. It does not guarantee finite-sample optimization or generalization of the neural model.

A.6 LAST-ITERATE CONVERGENCE

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (31)$$

Work throughout in the $\mathcal{D}$ -weighted geometry: pair functions of (x, y) by $\langle f, g \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}} \sum_y f(x, y) g(x, y)$, and let h be the *prompt-wise* negative entropy $h(\pi)(x) = \sum_y \pi(y | x) \log \pi(y | x)$, whose gradient at (x, y) is $1 + \log \pi(y | x)$. Its Bregman divergence in that geometry is exactly $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. The prompt expectation enters once, through the pairing, and is not also carried inside h . Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (32)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (32) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (33)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (35)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (36)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (35), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_{\text{ref}}). \quad (37)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (31).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (38)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because $\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (39)$$

Applying this inequality to $s = s(\pi_T)$ and using (31) proves (38). $\square$

B CONTROLLED BENCHMARK: CONSTRUCTION AND FULL RESULTS

Feasibility-preserving construction. For every seed we form a transitive base matrix and an odd-cardinality circulant skew-symmetric perturbation. Zero row sums keep the uniform reference disagreement invariant, while bisection selects the cycle magnitude so that $\rho^* = \max_\pi \min_k s_k(\pi)$ stays close to a target margin. We reject invalid probabilities rather than clip matrix entries. The resulting panel increases BT deviance from approximately 2.7×10^{-4} to 0.351 while retaining positive joint improvement in every instance. All policies are evaluated under the same adaptive-game values, not their method-native scales.

Robustness across construction and temperature. The primary construction is checked against an independently generated projected skew-symmetric family, $C = N(R - R^\top)N$ with N the symmetric projector annihilating both μ and a witness policy, and against a temperature sweep on that family. The trend replicates: adaptive-game rows stay flat while fixed-reference and scalar-reward rows fall. The effect *size* does not: the $\alpha = 1$ gap is +0.111 on the projected family against +1.172 on the circulant one, because the projected family’s BT deviance only reaches 0.174 against 0.351. We therefore claim robustness of the trend and do not attach the circulant effect size to it. Intervals in Table 7 are exact paired Student- t intervals over seeds, not bootstrap intervals.

C NATURAL SUPERVISION AND PREFERENCE MODELS

Dataset audit. Table 8 records the empirical role of each released dataset. ULTRAFEDBACK supplies complete per-completion ordinal attributes, so score-induced pairwise labels are transitive by construction. SAFERLHF supplies direct helpfulness and safety comparisons but almost exclusively isolated edges; zero observed triangles means the data cannot test within-objective cycles. It nevertheless contains substantial cross-objective disagreement and is therefore appropriate for the bargaining experiment.

Table 5: Full controlled results over five seeds under the common adaptive-game evaluator. Residuals are method-specific; the legacy map is a non-convergence ablation.

method	min surplus	sd	min s/ρ^*	sd	exploitability	inner resid.
$\alpha = 0.00, \rho^* = 0.0300, BT \text{ dev.} = 0.0003$						
NBPO direct	+0.0264	0.0011	0.881	0.037	0.0687	2×10^{-12}
Game-KS	+0.0268	0.0005	0.893	0.017	0.0685	10^{-9}
Game-util.	+0.0260	0.0014	0.866	0.048	0.0688	3×10^{-10}
Fixed-ref. Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-util.	+0.0260	0.0014	0.867	0.047	0.0689	0
legacy R -step [†]	+0.0252	0.0012	0.840	0.041	0.0675	5×10^{-10}
$\alpha = 0.25, \rho^* = 0.0300, BT \text{ dev.} = 0.0202$						
NBPO direct	+0.0269	0.0013	0.897	0.044	0.1091	10^{-7}
Game-KS	+0.0272	0.0007	0.905	0.025	0.1088	10^{-7}
Game-util.	+0.0265	0.0017	0.883	0.058	0.1094	4×10^{-8}
Fixed-ref. Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-util.	+0.0269	0.0015	0.895	0.049	0.1198	0
legacy R -step [†]	+0.0258	0.0014	0.860	0.046	0.1063	7×10^{-1}
$\alpha = 0.50, \rho^* = 0.0300, BT \text{ dev.} = 0.0813$						
NBPO direct	+0.0274	0.0013	0.914	0.044	0.1454	2×10^{-6}
Game-KS	+0.0276	0.0007	0.921	0.023	0.1450	2×10^{-7}
Game-util.	+0.0270	0.0017	0.900	0.057	0.1458	2×10^{-7}
Fixed-ref. Nash	+0.0243	0.0018	0.809	0.059	0.1945	0
BT-RM-Nash	+0.0243	0.0018	0.809	0.058	0.1945	0
BT-RM-util.	+0.0240	0.0019	0.800	0.064	0.1947	0
legacy R -step [†]	+0.0025	0.0044	0.082	0.147	0.1981	10×10^{-1}
$\alpha = 0.75, \rho^* = 0.0300, BT \text{ dev.} = 0.1882$						
NBPO direct	+0.0278	0.0013	0.926	0.042	0.1693	8×10^{-6}
Game-KS	+0.0278	0.0005	0.928	0.018	0.1692	8×10^{-7}
Game-util.	+0.0273	0.0017	0.909	0.056	0.1696	7×10^{-7}
Fixed-ref. Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-util.	+0.0137	0.0028	0.455	0.095	0.2724	0
legacy R -step [†]	-0.0453	0.0060	-1.511	0.201	0.2626	10×10^{-1}
$\alpha = 1.00, \rho^* = 0.0283, BT \text{ dev.} = 0.3510$						
NBPO direct	+0.0265	0.0033	0.931	0.041	0.1824	5×10^{-5}
Game-KS	+0.0264	0.0027	0.932	0.019	0.1825	3×10^{-6}
Game-util.	+0.0258	0.0037	0.908	0.063	0.1826	2×10^{-6}
Fixed-ref. Nash	-0.0063	0.0068	-0.241	0.277	0.3464	0
BT-RM-Nash	-0.0063	0.0068	-0.242	0.276	0.3464	0
BT-RM-util.	-0.0065	0.0070	-0.248	0.284	0.3465	0
legacy R -step [†]	-0.0949	0.0064	-3.373	0.406	0.3337	10^0

Table 6: Finite-pool solver audit at $\alpha = 1$ in the original controlled family. The direct practical solve recovers the positive-surplus proximal target; the deployed legacy map does not.

solution	min surplus	$/\rho^*$	inner residual	TV to exact prox.
Exact global Nash	+0.0158	+0.935	—	0.007
Exact proximal Nash	+0.0158	+0.936	—	0.000
NBPO direct inner solve	+0.0139	+0.821	2×10^{-6}	0.027
Legacy R -step map	-0.1851	-10.938	10^0	0.643
BT-RM-Nash	-0.1001	-5.918	0	0.495

Anti-symmetric general preference model. For objective k , a joint encoder produces $a_{\phi,k}(x, y, z)$ and we set

$$\ell_{\phi,k}(x, y, z) = \frac{1}{2} \{a_{\phi,k}(x, y, z) - a_{\phi,k}(x, z, y)\}, \quad P_{\phi,k}(y \succ z \mid x) = \sigma(\ell_{\phi,k}(x, y, z)).$$

This guarantees exact swap complementarity and self-ties without constraining the logit to a scalar reward difference. In the SafeRLHF diagnostic, both the joint model and the scalar BT baseline use RoBERTa-base so that encoder capacity and training cost are matched; this backbone choice is an experimental control rather than part of the NBPO method. Because the observed SafeRLHF graph contains no triangles, their near-null comparison is a calibration result rather than evidence that human preferences are transitive.

Table 7: Robustness across construction and temperature. Gaps in normalized worst surplus use paired Student- t 95% intervals across seeds. The temperature sweep uses the projected family.

cycle family	β	seeds	NBPO—FixedRef (t_{95})		NBPO—BT-RM (t_{95})		min ρ^*	converged
Circulant (primary)	0.25	5	+1.1724	[+0.8715, +1.4733]	+1.1733	[+0.8735, +1.4731]	0.0283	150/150
Projected skew-symmetric	0.25	5	+0.1114	[+0.0714, +0.1514]	+0.1115	[+0.0715, +0.1514]	0.0332	150/150
Projected skew-symmetric	0.10	3	+0.2381	[+0.0880, +0.3883]	+0.2385	[+0.0883, +0.3887]	0.0331	90/90
Projected skew-symmetric	0.50	3	+0.0448	[+0.0165, +0.0730]	+0.0449	[+0.0167, +0.0730]	0.0334	90/90

Table 8: Released-supervision audit. Zero observed human triangles means cyclicity is unidentifiable, not absent. Human conflict is cross-objective label disagreement on held-out pairs.

dataset	rows	prompts	responses	obs. triangles	human conflict	supervision	role
SAFERLHF	73,907	38,639	143,668	0	24.33%	direct pairwise	objective conflict
ULTRAFEEDBACK	63,967	63,967	4/prompt	n/a (score-induced)	n/a	ordinal attributes	transitive control

D TRAINING PROTOCOL AND BASELINES

The completed repaired study uses one outer stage and one training seed. The two losses share prompts, finite-pool targets, tokenizer, sequence masks, optimizer configuration, and policy-token exposure. Equal update counts do not imply equal FLOPs because regression also needs reference likelihoods. Table 11 records reported study dimensions and the announced repair configuration; execution-level verification remains required. Additional policy seeds and external baselines are unmeasured.

E SOLVER AND NEURAL-REALIZATION DIAGNOSTICS

The direct solve concerns a finite pool at a declared tolerance. It does not make the preference model, sampling approximation, or neural fit exact. Solver consistency, held-out regression, and generated-policy evaluation are reported separately.

Alternative realization. For comparison with (26), weighted behavior cloning uses

$$\mathcal{L}_{\text{WBC}}^{(t)}(\theta) = \mathbb{E}_x \sum_{i=1}^{n_x} p_{t,x}^*(i) [-\log \pi_\theta(y_{x,i} | x)]. \quad (40)$$

Weights are detached and full-vocabulary response log-probabilities are summed over tokens. With IID reference-policy occurrences, this weighted likelihood approximates population forward-KL projection; it remains a finite-sample realization rather than an exact population update (Agnihotri et al., 2026). Averaging $\frac{n_x}{2} [p_i^*(-\log \pi_\theta(y_i | x)) + p_j^*(-\log \pi_\theta(y_j | x))]$ over all unordered pairs equals the prompt-wise objective. Uniform teacher weights need not produce zero gradient on a finite sampled dataset.

Metrics and selection history. Write $t = \hat{h}_t^*$ and $h = h_t(\pi_\theta)$. We report $\text{nMSE} = \mathbb{E}[(h - t)^2] / \mathbb{E}[t^2]$, sign agreement, Pearson and Spearman correlations. Expectations use the stated split and pair measure. The zero predictor has $\text{nMSE} 1$. The reported oracle rescaling diagnostic is

$$a^* = \frac{\mathbb{E}[ht]}{\mathbb{E}[h^2]}, \quad \text{nMSE}^* = \min_{a \in \mathbb{R}} \frac{\mathbb{E}[(ah - t)^2]}{\mathbb{E}[t^2]}.$$

It fits a scalar on the evaluation observations and can reverse sign. It is neither centered explained variance nor a deployable calibration result. A direction-preserving version restricts $a \geq 0$; signed coefficients and this additional metric require the raw moments. If $\mathbb{E}[h^2] = 0$, set $a^* = 0$ and the residual to 1. Mean response log-ratio is measured under the sampled occurrence distribution; it is not total pool probability or a full-policy KL certificate.

The 1750-update comparison was specified first. The 250-update schedules were chosen after inspecting development trajectories and declared before their runs. Each uses its own warmup and cosine horizon. A checkpoint at step 250 of the long run is therefore a different condition. Diagnostic

Table 9: Held-out validation of checkpointed GPM and BT ensembles, seeds 41/42/43. Calibration uses validation data. Sparse comparison graphs cannot establish a cyclicity advantage.

model	objective	per-seed (mean of 3)			T	calibrated 3-seed ensemble				
		NLL	bal. acc	AUC		NLL	bal. acc	AUC	ECE	ens. sd
Anti-symmetric GPM	helpfulness	0.5543	0.7218	0.7999	1.282	0.5512	0.7257	0.8034	0.1228	0.0293
Anti-symmetric GPM	harmless.	0.5180	0.7556	0.8464	1.297	0.5131	0.7573	0.8512	0.1782	0.0357
Scalar BT	helpfulness	0.5560	0.7214	0.7974	1.270	0.5527	0.7250	0.8009	0.1220	0.0300
Scalar BT	harmless.	0.5212	0.7507	0.8431	1.265	0.5163	0.7544	0.8484	0.1779	0.0357

Table 10: Generated-pool pilot on 200 development prompts. Conflict and cycles are model predictions, not human-label measurements; cycles use a common comparator tournament.

geom.	dup. rate	entropy	ens. sd	GPM/BT sign	model conflict	pred. cycles	split-half r	ρ	sign agr.	dual evals	solve (s)
4+4	0.0073	0.6117	0.0429	0.9102	7.67%	0/800	0.9172	0.9132	0.8667	42	24.8
8+8	0.0167	0.6120	0.0414	0.9061	7.80%	10/11200	0.9242	0.9180	0.8925	30	18.4

thresholds were retained, but their use in development precludes describing the entire 2×2 as an untouched preregistered comparison.

Finite-pool distribution fidelity. For a candidate occurrence i , write $\Delta_i = \log \pi_\theta(y_i | x) - \log \pi_t(y_i | x)$ and reconstruct

$$\hat{p}_\theta(i | x) = \frac{p_{t,x}(i)e^{\Delta_i}}{\sum_j p_{t,x}(j)e^{\Delta_j}}.$$

This self-normalized importance distribution is evaluated against the solved target by forward KL on the fixed sampled support. It does not measure the full LLM divergence or probability outside that support. MSE-250 and WBC-250 reduce KL by 0.1675 and 0.1004, respectively; the reported intervals in Table 14 concern Pearson correlation, not KL improvement. Paired prompt intervals for KL remain to be reported. With uniform occurrence weights and $\Delta = 0$, target-argmax mass is algebraically $1/8$; reference top-1 agreement depends on tie-breaking.

Implementation history. The repaired study changed target construction, reference handling, tokenization, sampling, and training data jointly. Earlier failures cannot identify the effect of any single correction. Detailed debugging sweeps and the retired prompted-judge audit are retained in the code archive rather than used as performance evidence.

E.1 AGGREGATION-TARGET SENSITIVITY

On the reported pool, Nash weights (6.325, 6.186) yield targets close to equal weights with the same ℓ_1 mass. This is a target-level comparison; policy-level seed variation and statistical power are unknown. Weight asymmetry alone is not a sufficient criterion for benchmark informativeness because it can reflect objective units.

F EVALUATION PROTOCOL

Teacher-defined and independent evaluation. SafeRLHF surplus uses the frozen training preference models on held-out prompts, common comparator pools, and the same definition of $\hat{d}$. Greedy responses define point-mass decoders; they are not unbiased samples of the trained stochastic policy. Their positive surplus does not certify Algorithm 1 acceptance. Any stochastic-policy evaluation must sample both the trained policy and reference under a common declared protocol and account for prompt uncertainty.

IFEval, GSM8K, harmful-completion checks, and benign-refusal evaluation measure complementary behavior. AlpacaEval and Arena-Hard prompt sets are scored by a local reward model; these are not official leaderboard scores. Table 17 preserves the execution report’s Arena aggregate and token medians, whose subset and prompt scope remain unspecified. They do not populate subset-specific

Table 11: Reported study dimensions and announced repair settings. Optimizer, precision, and sampling settings must be reconciled with the execution manifest; the earlier 7,000-prompt plan is not a completed run.

Setting	Value
Base / adaptation	Llama-3.1-8B-Instruct / full fine-tuning
Train / dev / held-out prompts	2,000/500/1,000
Policy seeds / outer stages	one seed (ID pending manifest) / 1
Learner / comparator occurrences	8/8; independent streams
Sampling	temperature 1, top- $p = 1$; preserve terminal tokens
Preference models	frozen three-seed objective ensembles
Inner solve / neural target	direct finite-pool / canonical log-ratio
Optimizer / learning rate	AdamW / 5×10^{-7}
Adam moments / epsilon	(0.9, 0.95) / 10^{-8}
Schedule / declared updates	cosine / 1750 or 250
Warmup ratio / weight decay	0.10/0
Global batch / pair rows	32 pairs / 56,000
Prompt / total sequence limit	1024/2048 tokens
Precision / gradient clip	bf16 forward, fp32 reduction / 1
Opponent temperatures / proximal η	— (resolved config required)
Reference cache / optimizer-state checks	— (execution report required)
Model / split / config revisions	— (immutable manifest required)

Table 12: Matched mechanism controls. External PROSPER and MOPO comparisons require their actual algorithms and separate tuning contracts; game-maxmin or WBC alone is not their reproduction.

method	within-objective representation	aggregation	changed component relative to NBPO
NBPO	adaptive regularized game	Nash	—
Fixed-reference Nash	fixed reference margin	Nash	removes adaptive opponent
BT-RM-Nash	scalar BT reward	Nash	imposes scalar reward representation
Game-utilitarian	adaptive regularized game	equal-weight sum	replaces Nash aggregation
Game-KS	adaptive regularized game	normalized proportional gains	alternative bargaining rule
BT-RM-utilitarian	scalar BT reward	equal-weight sum	scalar representation and aggregation control

main-table cells. The exact IFEval strict metric and numerical confidence intervals must be reconciled with evaluator outputs. A confidence interval containing 0.5 does not establish win-rate equivalence.

Refusal concentration and teacher truncation. The execution report identifies 448 of 1,000 prompts with the same short base refusal. Its refusal percentages require explicit denominators. This output-defined, post-hoc group shows larger teacher gains than prompts with substantive base answers, but adding alternatives to refusals can benefit several training objectives. It is not specific evidence for Nash aggregation. A reported 32% teacher-truncation rate also requires the truncation unit and policy to be specified; the measured surplus concerns the implemented evaluator. Length-controlled correlation does not resolve omitted response content or reward-model length bias. An independent full-response audit should stratify by base refusal and teacher truncation, with benign refusal evaluated separately. Raising a teacher’s sequence limit beyond its supported context is not a valid correction.

F.1 EMPIRICAL PARETO EVALUATION

The population Nash optimum is Pareto efficient under the stated assumptions. A single proximal step and an approximate neural realization do not inherit that certificate. For the planned comparison, canonical NBPO is one policy point; scalarized DPO baselines (Rafailov et al., 2023) at weights $\{0, 0.25, 0.5, 0.75, 1\}$ provide endpoints and intermediate compromises. Actual MOPO and PROSPER adaptations require their own algorithms and a declared tuning budget. The corresponding results are unmeasured in this snapshot.

Report independent helpfulness and harmlessness outcomes on both the entire frozen test set and a subset defined by disagreement between human objective labels. Keep training-teacher game surplus

Table 13: All recorded development evaluations of the repaired runs. The first column identifies the declared schedule; “Updates” is the checkpoint step. Oracle nMSE* permits signed rescaling. No trajectory point is substituted for its run’s declared final checkpoint.

Projection	Updates	nMSE	nMSE*	Sign acc.	Pearson	Spearman	$\overline{\log(\pi_\theta/\pi_\epsilon)}$
NBPO-MSE, 1750	250	0.839	0.734	0.692	0.516	0.499	-1.13
	500	1.226	0.796	0.688	0.452	0.465	-1.86
	750	1.482	0.824	0.678	0.419	0.452	-2.42
	1000	1.568	0.839	0.694	0.401	0.449	-2.38
	1250	1.517	0.836	0.688	0.404	0.450	-2.27
	1500	1.432	0.831	0.691	0.412	0.455	-2.12
	1750	1.418	0.828	0.693	0.415	0.460	-2.07
NBPO-WBC, 1750	250	0.920	0.824	0.645	0.419	0.415	-0.75
	500	1.932	0.938	0.623	0.250	0.297	-2.04
	750	3.776	0.995	0.569	0.074	0.124	-3.27
	1000	6.121	0.999	0.535	-0.028	0.018	-4.39
	1250	11.611	0.979	0.496	-0.143	-0.096	-6.15
	1500	12.672	0.979	0.489	-0.144	-0.097	-6.63
	1750	13.070	0.977	0.486	-0.152	-0.105	-6.68
NBPO-MSE, 250 [†]	50	0.737	0.702	0.675	0.546	0.507	-0.77
	100	0.751	0.674	0.691	0.571	0.537	-1.11
	150	0.710	0.659	0.695	0.584	0.548	-1.03
	200	0.701	0.668	0.693	0.576	0.539	-0.87
	250	0.690	0.655	0.702	0.588	0.552	-0.90
NBPO-WBC, 250 [†]	50	0.866	0.839	0.628	0.401	0.388	-0.39
	100	0.807	0.777	0.655	0.473	0.450	-0.55
	150	0.755	0.730	0.678	0.520	0.499	-0.64
	200	0.770	0.743	0.682	0.507	0.497	-0.57
	250	0.780	0.758	0.687	0.492	0.488	-0.51

Table 14: Independent reconstruction on the development pool. Correlation intervals are prompt-clustered. KL improvement is reference minus policy KL. Target-argmax mass is $\hat{p}_\theta(i^*)$, $i^* \in \arg \max_i p^*(i)$; tie rules must be fixed. The WBC-1750 improvement is reported at source precision and differs by 10^{-4} from subtracting rounded KL entries.

Policy	Pearson [clustered CI]	$D_{KL} \downarrow$	KL improvement [†]	Top-1	Argmax mass
Base	—	0.7879	+0.0000	0.108	0.1250
MSE (1750)	+0.417 [0.370, 0.463]	1.2131	-0.4252	0.420	0.3135
WBC (1750)	-0.150 [-0.205, -0.095]	6.0634	-5.2754	0.124	0.1113
MSE (250) [†]	+0.589 [0.554, 0.622]	0.6204	+0.1675	0.406	0.2906
WBC (250) [†]	+0.496 [0.459, 0.531]	0.6875	+0.1004	0.362	0.2283

separate from these external axes. A policy is empirically nondominated only relative to the evaluated comparison set and evaluator; this is not global optimality over LLM policies. Show individual policy seeds, paired prompt uncertainty, and all declared weight settings. Curves joining measured points are visual guides, not evaluated mixture policies. Nash welfare is reported only when every surplus is positive; nonpositive cases are marked infeasible rather than clipped. Any hypervolume normalization and reference point must be fixed before final results.

Uncertainty and provenance. Report every policy seed, then mean and standard deviation across training seeds. Paired prompt bootstrap intervals condition on a checkpoint and do not include training-seed variability. Freeze development choices before confirmatory comparisons and retain the history of benchmark exposure. Current numerical entries were supplied for one seed; dashes are unmeasured, not zeros. All result updates must record checkpoint, split, decoding, evaluator, sample count, and source artifact. Unmatched literature scores do not fill matched-baseline rows.

Table 15: Measured direct-solver scaling by prompt count, pool size, and CPU workers. Times cover finite-pool solving and artifact generation, not LLM rollout or training.

prompts	pool	workers	solve (s)	dual evals	s/eval	proj. KKT	inner resid.	peak RSS (MB)	artifact (s)
200	4+4	32	1.5	31	0.047	3×10^{-14}	3×10^{-11}	467	0.01
200	8+8	32	1.7	35	0.049	4×10^{-15}	5×10^{-12}	498	0.02
200	16+16	32	1.8	32	0.055	2×10^{-14}	10^{-11}	499	0.02
1000	4+4	96	4.4	33	0.132	7×10^{-15}	2×10^{-11}	499	0.03
1000	8+8	96	3.2	26	0.122	4×10^{-15}	6×10^{-12}	504	0.04
1000	16+16	96	4.4	37	0.119	3×10^{-13}	10×10^{-11}	534	0.07
7000	4+4	96	10.6	28	0.379	7×10^{-10}	4×10^{-11}	534	0.11
7000	8+8	96	10.9	27	0.404	2×10^{-15}	3×10^{-11}	544	0.22
7000	16+16	96	15.3	30	0.508	10^{-14}	7×10^{-11}	633	0.39

Table 16: Nash versus mass-matched utilitarian target diagnostics on the reported pool. Policy comparisons remain unmeasured.

Diagnostic	Value
Mean target TV	0.0008
Maximum target TV	0.0075
Pairwise target difference RMS	0.0034
Target RMS on the aggregation panel	1.125
Nash λ_{help}	6.325
Nash λ_{safe}	6.186

Table 17: Completed downstream snapshot supplied by the execution report. Arena is an unassigned local-proxy aggregate; token medians have an unassigned prompt scope. These values are preserved without relabeling them as hard/creative or Alpaca-specific results. All policies currently have one training seed.

Policy	$\min_k s_k$	IFEval	GSM8K	Harm↓	Alpaca	Arena (unassigned)	Tokens (unassigned)
Base	-0.0405	.752	.867	.281	.500	.500	457
MSE (1750)	+0.1337	.758	.864	.284	.419	.435	481
WBC (1750)	+0.0738	.741	.852	.256	.578	.594	457
MSE (250) [†]	+0.1186	.760	.870	.269	.497	.516	460
WBC (250) [†]	+0.0569	.741	.867	.266	.486	.542	463

Table 18: Evaluation contracts and fields to complete from actual artifacts. These counts specify the requested full evaluation sets, not a claim that every job has completed.

Set / target count	Metric and completion contract
SafeRLHF / 1000	Teacher-defined objective values; comparator and disagreement hashes; greedy and stochastic evaluations kept separate.
IFEval / 541	Official strict prompt and instruction accuracy; loose scores secondary. Verify the reported strict column against evaluator output.
GSM8K / 1319	Fixed zero-shot-CoT prompt and answer extractor; exact match and parse-failure count.
HarmBench / text test	Report actual category counts and classifier version. Use category-specific hashing where required.
XSTest / 250 safe, 200 unsafe	Benign refusal and unsafe-request handling separately; report classifier or blinded manual assessment.
Alpaca / 805	Same-prompt local reward-model win rate versus base, ties half; response length from the same prompt set.
Arena / 500 hard, 250 creative	Report subset-specific local proxy win rates; an unassigned aggregate does not populate these columns.
MT-Bench / 80	Optional two-turn responses; no official score without the stated judging protocol.